



Advanced Time Series Analysis and Forecasting

Chapter 9: VaR, ES and backtesting: elicibility, scoring and model risk



Daniel Traian PELE

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material (1/3)

- ▣ **ACI**: Adaptive Conformal Inference
- ▣ **ADAPT**: Adaptive (CAViaR specification)
- ▣ **AI**: Artificial Intelligence
- ▣ **AR**: AutoRegressive (model); in event studies: Abnormal Return
- ▣ **ARFIMA**: AutoRegressive Fractionally Integrated Moving Average (model)
- ▣ **ARMA**: AutoRegressive Moving Average (model)
- ▣ **ARMA-GARCH**: ARMA mean with GARCH variance
- ▣ **AS**: Asymmetric Slope (CAViaR specification)
- ▣ **BET**: Bucharest Exchange Trading (index)
- ▣ **BIC**: Bayesian Information Criterion
- ▣ **BNR**: Banca Națională a României (the National Bank of Romania)
- ▣ **BUX**: Budapesti Értéktőzsde index (the Budapest Stock Exchange index)
- ▣ **DM**: Diebold–Mariano (test)
- ▣ **DQ**: Dynamic Quantile (test of Engle and Manganelli)
- ▣ **EDF**: Empirical Distribution Function
- ▣ **EM**: Engle and Manganelli (2004), CAViaR
- ▣ **EODHD**: EOD Historical Data (market-data provider)
- ▣ **ES**: Expected Shortfall
- ▣ **EVT**: Extreme Value Theory
- ▣ **EWMA**: Exponentially Weighted Moving Average
- ▣ **FHS**: Filtered Historical Simulation
- ▣ **FZ**: Fissler–Ziegel (class of scoring functions for VaR and ES)
- ▣ **FZ-1F**: One-factor GAS model for VaR and ES (label of Patton, Ziegel and Chen)
- ▣ **FZ-2F**: Two-factor GAS model for VaR and ES (label of Patton, Ziegel and Chen)

Acronyms used in this material (2/3)

- ▣ **FZ0**: Zero-homogeneous Fissler–Ziegel loss (Patton, Ziegel and Chen)
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ **GARCH-EDF**: GARCH with innovations from the empirical distribution function
- ▣ **GARCH-FZ**: GARCH model estimated by FZ0 minimisation
- ▣ **GARCH-N**: GARCH with Normal innovations
- ▣ **GAS**: Generalised Autoregressive Score (model)
- ▣ **GAS-1F**: One-factor GAS model for VaR and ES
- ▣ **GAS-2F**: Two-factor GAS model for VaR and ES
- ▣ **GAS-FZ**: GAS models for VaR and ES driven by the FZ0 loss
- ▣ **GCH**: GARCH (short label in tables)
- ▣ **GCH-EDF**: ARMA-GARCH with innovations from the empirical distribution function
- ▣ **GCH-FZ**: GARCH model estimated by FZ0 minimisation (table label)
- ▣ **GCH-N**: ARMA-GARCH with Normal innovations
- ▣ **GJR**: Glosten–Jagannathan–Runkle (asymmetric GARCH)
- ▣ **GJR-GARCH**: Glosten–Jagannathan–Runkle GARCH
- ▣ **GM**: General Motors (stock)
- ▣ **GPL**: Generalised Piecewise Linear (scoring function)
- ▣ **GW**: Giacomini–White (test of conditional predictive ability)
- ▣ **HAC**: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ **HC0**: Heteroskedasticity-Consistent covariance of White (type 0)
- ▣ **HS**: Historical Simulation
- ▣ **IBM**: International Business Machines (stock)

Acronyms used in this material (3/3)

- ▣ **IG**: Indirect GARCH (CAViaR specification)
- ▣ **LLM**: Large Language Model
- ▣ **LR**: Likelihood Ratio (test)
- ▣ **MA**: Moving Average
- ▣ **MAR33**: Basel Framework, chapter MAR33 (internal models, capital requirements)
- ▣ **MCS**: Model Confidence Set
- ▣ **MFM**: Modelling Financial Markets
- ▣ **MLE**: Maximum Likelihood Estimation
- ▣ **MSE**: Mean Squared Error
- ▣ **PIT**: Probability Integral Transform
- ▣ **PX**: Prague Stock Exchange index (index Burzy cenných papírů Praha)
- ▣ **PZC**: Patton, Ziegel and Chen (2019)
- ▣ **QLIKE**: Quasi-LIKElihood loss
- ▣ **QML**: Quasi-Maximum Likelihood
- ▣ **RQ**: Regression Quantile (criterion of Engle and Manganelli)
- ▣ **RW**: Random Walk
- ▣ **RW-125**: Rolling Window of 125 days (historical simulation)
- ▣ **RW-250**: Rolling Window of 250 days (historical simulation)
- ▣ **RW-500**: Rolling Window of 500 days (historical simulation)
- ▣ **SAV**: Symmetric Absolute Value (CAViaR specification)
- ▣ **TSA**: Time Series Analysis
- ▣ **VAR**: Vector AutoRegression
- ▣ **VIX**: CBOE Volatility Index
- ▣ **WIG20**: Warszawski Indeks Giełdowy 20 (the Warsaw Stock Exchange index of the 20 largest companies)

Today's question and route

- **Question:** a VaR or an ES forecast is a point forecast of a tail functional; how do we estimate dynamic models for it, test whether it is calibrated and decide which forecast is better?
 - ▶ the tail is observed rarely: a forecast must be judged with a loss that is consistent for its target, on few informative days
- **Route** of the chapter
 - ▶ risk measures as functionals; elicibility, the Fissler–Ziegel class, FZ0 and Murphy diagrams
 - ▶ quantile regression and CAViaR; semiparametric (VaR, ES) models and joint regression
 - ▶ backtesting as conditional calibration; estimation risk in backtests; comparison with the MCS in stress periods
 - ▶ the horizon (square-root-of-time), model risk, extremes under dependence, conformal calibration
- We build on [Chapter 1](#) (scoring rules, elicibility, DM, MCS), [Chapter 0](#) (HAC, bootstrap), [Chapter 8](#) (volatility) and [TSA, Chapter 5](#) (GARCH); Seminar 9 comes before this lecture

Learning outcomes

- ▣ Characterise the scoring functions consistent for quantiles and for the pair (VaR, ES), and prove that ES alone is not elicitable
- ▣ Estimate CAViaR models with the Engle–Manganelli procedure, compute their standard errors and run the DQ test
- ▣ Estimate semiparametric dynamic (VaR, ES) models by FZ0 minimisation and replicate the out-of-sample comparison of Patton, Ziegel and Chen
- ▣ Test conditional calibration with identification functions, account for estimation risk, and compare risk models with DM tests, Murphy diagrams and the MCS
- ▣ Quantify horizon, model and estimation risk, and correct a miscalibrated VaR by adaptive conformal inference

Reading, data and tools

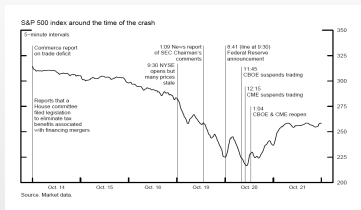
- Theory: [Gneiting \(2011\)](#); [Fissler and Ziegel \(2016\)](#); [Ehm et al. \(2016\)](#); [Nolde and Ziegel \(2017\)](#)
 - ▶ Models: [Koenker and Bassett \(1978\)](#); [Engle and Manganelli \(2004\)](#); [Patton, Ziegel and Chen \(2019\)](#); [Dimitriadis and Bayer \(2019\)](#)
 - ▶ Survey of forecasting practice: [Petropoulos et al. \(2022\)](#)
- Python Quantlets of this chapter: [Quantlets/Ch_09](#)
 - ▶ CAViaR, the GAS-FZ models, the joint regression, the backtests, DM and the MCS written in `numpy/scipy`; `numba` only speeds up the recursions
- Lecture notebook: [open in Google Colab](#)

Data used in this chapter

Series	Source, sample	In sample / out of sample
S&P 500, DAX	EODHD daily closes, January 1990 – September 2026	1990–1999 / 2000–2026
BET EUR/RON	EODHD, official closes, 2000–2026 BNR reference rate, July 2005 – September 2026	2000–2009 / 2010–2026 to June 2015 / from July 2015
Bitcoin	EODHD, 7 days a week, September 2014 – September 2026	2014–2019 / 2020–2026
VIX	EODHD daily closes	regressor of the joint (VaR, ES) regression

- Daily log returns in %; ten in-sample years as in Patton, Ziegel and Chen (2019), five for Bitcoin; stress periods: September 2008 – March 2009, February – June 2020, 2022, March – June 2025

From a crash to a regulatory functional



The S&P 500 around 19 October 1987

Figure: Mark Carlson, Federal Reserve Board (2006); public domain; Wikimedia Commons

- 1987: a one-day fall of about 20% in US equities shows that the tail, not the variance, decides solvency
- 1996: the Basel Committee adopts VaR for market risk and a traffic-light backtest of its hits [Basel Committee \(1996\)](#)
- 1999–2002: VaR is not coherent; ES is [Artzner et al. \(1999\)](#), [Acerbi and Tasche \(2002\)](#)
- 2011–2016: ES is not elicitable [Gneiting \(2011\)](#), but (VaR, ES) is jointly elicitable [Fissler and Ziegel \(2016\)](#)
- FRTB: capital on ES 2.5%, backtests on VaR 1% and VaR 2.5% [Basel Committee \(MAR33\)](#)

Basel: the forecaster and the referee



Bank for International Settlements, Basel, 2018

Photo: Michal Pleskowicz (2018); CC BY-SA 4.0;

[Wikimedia Commons](#)

- A bank issues a daily forecast of its own tail; the supervisor checks it after the fact
- Two questions that statistics separates: is the forecast **calibrated** (backtest) and is it **better** than another (comparison)?
- Elicitability decides which forecasts can be compared at all [Nolde and Ziegel \(2017\)](#)
- The incentive matters: a scoring rule rewards honest reporting only if it is strictly consistent for the target

Known from TSA and MFM, and new here

- Known: VaR and ES by historical simulation, parametric, filtered HS and EVT; Kupiec and Christoffersen tests; the Basel traffic light (TSA, Chapter 5; MFM, Chapters 7–8)
- New: the forecasting view
 - ▶ which losses are consistent for a quantile and for (VaR, ES); what elicibility buys and what it does not
 - ▶ M-estimation of dynamic quantile and ES models: identification, asymptotics, non-smooth objectives
 - ▶ backtests as moment tests of conditional calibration, with estimation risk; comparisons that survive multiple testing
- Replications: Engle and Manganelli (2004) and Patton, Ziegel and Chen (2019), with the specifications of the papers, on our data

VaR and ES as functionals (1/2): definitions

- VaR is minus a quantile of the return distribution

$$q_{\alpha}(F) = \inf\{y : F(y) \geq \alpha\}, \quad \text{VaR}_{\alpha}(F) = -q_{\alpha}(F)$$

- ▶ Y : the return, with distribution function F ; α : the level, i.e. the target hit rate (VaR 1%: $\alpha = 0.01$)
- ▶ $q_{\alpha}(F)$: the smallest y with $F(y) \geq \alpha$; a correct VaR is exceeded with probability $\Pr(Y < -\text{VaR}_{\alpha}) = \alpha$

- ES is minus the average of the quantiles below level α [Acerbi and Tasche \(2002\)](#)

$$\text{ES}_{\alpha}(F) = -\frac{1}{\alpha} \int_0^{\alpha} q_u(F) du$$

- ▶ for continuous F : $\text{ES}_{\alpha} = -\mathbb{E}[Y \mid Y \leq q_{\alpha}]$, minus the mean return on the worst α share of days
- ▶ ES is coherent (subadditive) [Artzner et al. \(1999\)](#); VaR is not; ES needs $\mathbb{E}|Y| < \infty$

VaR and ES as functionals (2/2): conventions

- In formulas we work on the return scale with the pair $(v, e) = (q_\alpha, -ES_\alpha)$
 - ▶ v : the α -quantile of returns; e : the mean return beyond it; for the usual levels both are negative and $e \leq v < 0$
- Both are **functionals** $T(F)$: maps from a distribution to a number
 - ▶ a forecast of them is a point forecast of the predictive distribution, as in [Chapter 1](#)

Conditional risk forecasts: three routes (1/2)

- Target: the functional of the conditional distribution

$$(v_t, e_t) = T(F_{t|t-1})$$

- ▶ $F_{t|t-1}$: distribution of the return Y_t given $\mathcal{F}_{t-1}$, the information up to day $t - 1$

- **Location–scale**: the return is a mean plus a volatility times an i.i.d. shock

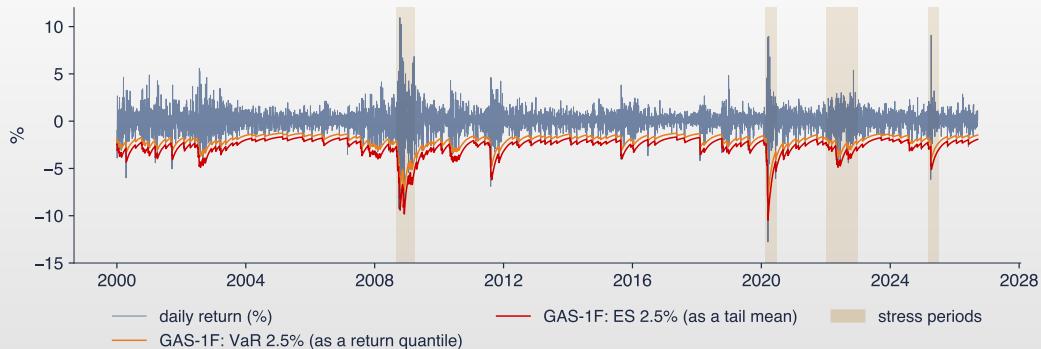
$$Y_t = \mu_t + \sigma_t \eta_t \Rightarrow v_t = \mu_t + \sigma_t q_\alpha(\eta), \quad e_t = \mu_t + \sigma_t \mathbb{E}[\eta \mid \eta \leq q_\alpha(\eta)]$$

- ▶ μ_t, σ_t : conditional mean and volatility (ARMA, GARCH); $q_\alpha(\eta)$: quantile of the shock distribution
- ▶ GARCH with Normal, skewed-t [Hansen \(1994\)](#) or empirical (EDF) innovations; the ratio e_t/v_t is constant over time when $\mu_t = 0$

Conditional risk forecasts: three routes (2/2)

- **Direct quantile dynamics** Engle and Manganelli (2004): $v_t = v(\mathcal{F}_{t-1}; \beta)$
 - ▶ $v(\cdot; \beta)$: a recursion with parameters β , estimated by the quantile (pinball) loss, without a distribution
- **Joint semiparametric**: $(v_t, e_t) = (v, e)(\mathcal{F}_{t-1}; \theta)$
 - ▶ two recursions with parameters θ , estimated by a Fissler–Ziegel loss Patton, Ziegel and Chen (2019), Dimitriadis and Bayer (2019)
- Every route produces a point forecast of a functional; the scoring function of the next section is what makes them comparable

Thirty-six years of tail forecasts



- S&P 500 daily returns; the GAS-1F model of Patton, Ziegel and Chen, parameters estimated once on 1990–1999 and then only filtered; shaded: the four stress periods

Interpreting the long-run forecasts

- ▣ The ES 2.5% forecast ranges from about -2.44% (median) to -10.5% on 17 March 2020: tail risk moves as much as volatility
- ▣ The forecasts react only on hit days and decay smoothly otherwise: the score-driven update of the GAS model
- ▣ Parameters fixed in 1999 still track 2008, 2020 and 2025: the dynamics generalise, the level does not need re-estimation
- ▣ Whether this is good enough is a testing question (Sections 5–6), not a visual one

Recap: Risk measures as functionals

- ▣ VaR is a quantile, ES an integral of quantiles; both are functionals of the predictive distribution
- ▣ Three routes: location–scale, direct quantile dynamics, joint semiparametric models
- ▣ The convention: VaR 1%, ES 2.5%, α is the target hit rate

Consistency, elicibility, identification (1/2)

- A scoring function $S(x, y)$ assigns a loss to the forecast x when y is observed; it is **consistent** for T on a class $\mathcal{F}$ if [Gneiting \(2011\)](#)

$$\mathbb{E}_F S(T(F), Y) \leq \mathbb{E}_F S(x, Y) \quad \text{for all } x \text{ and } F \in \mathcal{F}$$

- ▶ $\mathbb{E}_F$: expectation when $Y \sim F$; the true functional value has the smallest expected loss
- ▶ **strictly** consistent: equality forces $x = T(F)$; T is **elicitable** if a strictly consistent S exists ([Chapter 1](#))

Consistency, elicibility, identification (2/2)

- ▣ **Identification function:** a function $V(x, y)$ whose mean is zero exactly at the true value

$$\mathbb{E}_F V(x, Y) = 0 \iff x = T(F)$$

► quantile: $V(x, y) = 1\{y \leq x\} - \alpha$, with $1\{\cdot\}$ the indicator (1 if true, 0 otherwise); mean: $V(x, y) = x - y$

- ▣ Osband's principle: the slope of the expected score is a positive multiple of the expected identification function
Fissler and Ziegel (2016)

$$\partial_x \mathbb{E}_F S(x, Y) = h(x) \mathbb{E}_F V(x, Y), \quad h > 0$$

► scores are integrated identification functions

- ▣ Identification functions give **backtests** (is the forecast calibrated?); scoring functions give **comparisons** (which forecast is better?)
Nolde and Ziegel (2017)

Quantiles: all consistent scores

- ▣ **Theorem Gneiting (2011):** under mild conditions, S is consistent for q_α if and only if it is generalised piecewise linear (GPL)

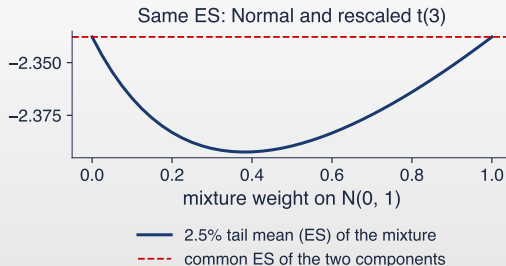
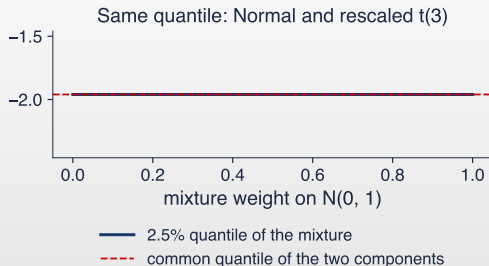
$$S(x, y) = (1\{y \leq x\} - \alpha)(G(x) - G(y)), \quad G \text{ non-decreasing}$$

- ▶ $G(x) = x$: the pinball loss; G strictly increasing gives strict consistency; $G(x) = \ln x$ for positive variables gives a scale-free (zero-homogeneous) score
- ▶ why: $\partial_x \mathbb{E}_F S(x, Y) = G'(x)(F(x) - \alpha)$, negative below q_α , positive above ([Appendix](#))
- ▣ Different G give the same optimal forecast but can rank two misspecified forecasts differently: the choice of G is a modelling choice
- ▣ Expectiles (asymmetric squared loss) are the only elicitable coherent risk measures [Ziegel \(2016\)](#); [Taylor \(2008\)](#) uses them for VaR and ES

ES is not elicitable

- **Necessary condition** [Gneiting \(2011\)](#): if T is elicitable, its level sets are convex
 - ▶ level set: $\{F : T(F) = t\}$, all distributions with the same functional value t ; convex: closed under mixtures $\lambda F_0 + (1 - \lambda)F_1$, $\lambda \in [0, 1]$
 - ▶ proof: if $\mathbb{E}_{F_0} S(t, Y) \leq \mathbb{E}_{F_0} S(x, Y)$ and the same for F_1 , the inequality holds for the mixture, because expectations are linear in F
- Quantiles pass: if $F_0(t) = F_1(t) = \alpha$, then every mixture has $F_\lambda(t) = \alpha$
- ES fails: the mixture changes the quantile, so it changes which part of each component enters the tail mean [Weber \(2006\)](#)
 - ▶ consequence: no loss function can rank ES forecasts alone; average losses of ES forecasts are not meaningful by themselves
- Variance fails for the same reason; (mean, second moment) and (VaR, ES) are elicitable as pairs ([Fissler and Ziegel \(2016\)](#), “higher order elicibility”)

Level sets under mixing



- Left: $N(0, 1)$ and a $t(3)$ rescaled to the same 2.5% quantile; right: $N(0, 1)$ and a $t(3)$ rescaled to the same ES 2.5%; curves: the functional of the mixture

Interpreting the level sets

- ▣ The quantile of every mixture stays at -1.96 : the level set of $q_{0.025}$ is convex
- ▣ Both components have tail mean -2.338 ; the mixture moves away by up to 0.055 (at weight 0.38): the level set of ES is not convex
- ▣ The gap is small in numbers but decisive in logic: one counterexample rules out every strictly consistent loss for ES alone
- ▣ The way out is to forecast the quantile together with ES: the pair has a convex level set

Joint elicibility: the Fissler–Ziegel class (1/2)

- **Theorem** Fissler and Ziegel (2016), in the form of Patton, Ziegel and Chen (2019), eq. (4): for $e \leq v$, the loss

$$S(v, e, y) = (1\{y \leq v\} - \alpha)(G_1(v) - G_1(y)) + G_2(e) \left(v - e + \frac{1}{\alpha} 1\{y \leq v\}(y - v) \right) - \mathcal{G}_2(e)$$

- ▶ is consistent for $(q_\alpha, -\text{ES}_\alpha)$ if G_1 is non-decreasing and $G_2 = \mathcal{G}_2'$ is positive and increasing; strictly so under mild conditions
- Reading the two parts
 - ▶ first term: a GPL quantile score for v (previous slides); G_1 : any non-decreasing function
 - ▶ second part: a Bregman-type score for e , evaluated against the tail mean implied by v ; $\mathcal{G}_2$: an antiderivative of G_2

Joint elicibility: the Fissler–Ziegel class (2/2)

- Identification function of the pair: one component for the quantile, one for the tail mean

$$V_1 = 1\{y \leq v\} - \alpha, \quad V_2 = e - v + \frac{1}{\alpha} 1\{y \leq v\}(v - y)$$

- ▶ $\mathbb{E}V_1 = 0$: v is the α -quantile; $\mathbb{E}V_2 = 0$ at the true quantile is the Acerbi–Tasche formula
- ▶ ES is the tail mean only together with the right quantile
- Consequence [Fissler, Ziegel and Gneiting \(2016\)](#): ES can be **compared** through the pair; traditional ES backtests still need more than ES (Section 5)

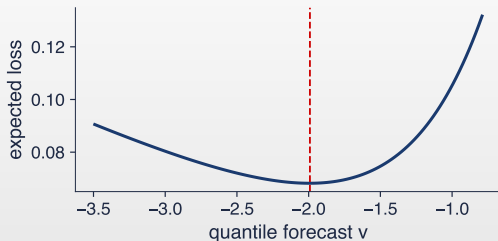
FZ0: the zero-homogeneous member

- Patton, Ziegel and Chen (2019), Proposition 1: with $v, e < 0$, loss **differences** are invariant to rescaling Y if and only if $G_1 = 0$ and $G_2(e) = -1/e$

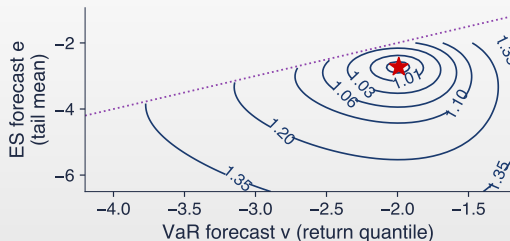
$$L_{\text{FZ0}}(y, v, e; \alpha) = -\frac{1}{\alpha e} 1\{y \leq v\}(v - y) + \frac{v}{e} + \ln(-e) - 1$$

- ▶ first term: positive only on hit days ($y \leq v$), proportional to the size of the hit; $v/e \in (0, 1]$; $\ln(-e)$ penalises an exaggerated ES
- ▶ the VaR part resembles the pinball loss; the ES part resembles QLIKE (Chapter 8)
- Why zero homogeneity matters for time series
 - ▶ with a non-homogeneous score, volatile days dominate the average loss and the DM test (heteroskedastic loss differences)
 - ▶ FZ0 ranks forecasts in % and in basis points identically; its value can be negative (EUR/RON)
 - ▶ proof of consistency and of homogeneity: ▶ [Appendix](#)
- Iso-expected-loss contours are convex under mild conditions, which helps numerical minimisation Patton, Ziegel and Chen (2019)

Expected losses around the truth



— expected pinball loss
 - - - true quantile = -1.99



..... $e = v$ (boundary)
 ★ true (quantile, tail mean) = (-1.99, -2.73)

- Student $t(5)$ with unit variance, $\alpha = 0.025$, expectations by simulation (4×10^5 draws): left, expected pinball loss; right, contours of the expected FZ0 loss

Interpreting the expected losses

- ▣ The pinball curve is minimal at -1.98 , the true quantile -1.99 : consistency in action
- ▣ The FZ0 surface is minimal at $(-1.99, -2.73)$, minimum 1.004; contours are convex and elongated
- ▣ Both surfaces are flat near the optimum: large forecast differences cost little expected loss, so tests need many tail days
- ▣ Forecasts with $e > v$ are inadmissible: estimation must enforce the ordering

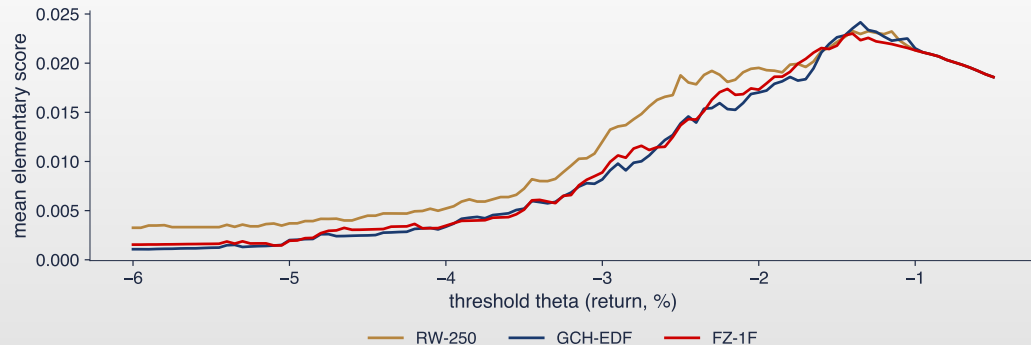
Murphy diagrams: ranking for all consistent scores

- Ehm et al. (2016): every GPL quantile score is a mixture of **elementary scores**

$$S_{\theta}(x, y) = (1\{y < x\} - \alpha)(1\{\theta < x\} - 1\{\theta < y\}), \quad S(x, y) = \int S_{\theta}(x, y) dH(\theta)$$

- ▶ θ : a threshold; S_{θ} : the loss of a binary decision “is the return below θ ?”; H : a non-decreasing weight over thresholds, corresponding to G
- **Murphy diagram**: the average $\bar{S}_{\theta}$ of each forecast, plotted against θ
 - ▶ forecast A dominates B for every consistent score if and only if its curve is below for all θ
 - ▶ crossing curves: the ranking depends on G , i.e. on which thresholds the user cares about
- A check on the pinball ranking that costs one line of code

A Murphy diagram for VaR 2.5%



- S&P 500, out of sample 2000–2016 (PZC design); mean elementary scores of three VaR 2.5% forecasts against the threshold θ

Interpreting the Murphy diagram

- ▣ GARCH-EDF lies below RW-250 at 91% of the thresholds: the dynamic forecast dominates almost uniformly
- ▣ GAS-1F and GARCH-EDF cross: GAS-1F is better at only 38% of the thresholds; average pinball 7.66 and 7.45 ($\times 10^{-2}$)
- ▣ The two good models differ in deep thresholds (below -4%), where data are scarce
- ▣ A claim that “model A is better” should say for which scores; the diagram answers it

Recap: Elicitability

- ▣ Scores compare, identification functions test; both come from the same theory
- ▣ Quantiles: GPL scores; ES: not alone; (VaR, ES): the Fissler–Ziegel class
- ▣ FZ0 is scale-free; Murphy diagrams check rankings for all consistent scores

Quantile regression as M-estimation (1/2)



Roger Koenker, Oberwolfach, 2012

Photo: Ivonne Vetter, MFO (2012); CC

BY-SA 2.0 de; Wikimedia Commons

- **Koenker and Bassett (1978)**: the α -quantile of y_t given x_t is modelled as $x_t'\beta$ and estimated by minimising the pinball loss

$$\hat{\beta}(\alpha) = \arg \min_{\beta} \sum_t \rho_{\alpha}(y_t - x_t'\beta), \quad \rho_{\alpha}(u) = u(\alpha - 1\{u < 0\})$$

- ▶ x_t : vector of regressors; ρ_{α} : the check (pinball) function; the problem is a linear program
- ▶ ρ_{α} has slope α for positive residuals and $\alpha - 1$ for negative ones
- ▶ first-order condition: $\sum_t x_t(1\{y_t \leq x_t'\hat{\beta}\} - \alpha) \approx 0$, the identification function

Quantile regression as M-estimation (2/2)

- Asymptotic distribution [Koenker \(2005\)](#): a sandwich in which the density at the quantile appears

$$\sqrt{T}(\hat{\beta} - \beta) \rightarrow N(0, \alpha(1 - \alpha)D^{-1}\Omega D^{-1}), \quad \Omega = \mathbb{E}[x_t x_t'], \quad D = \mathbb{E}[f_t(q_t)x_t x_t']$$

- ▶ $f_t(q_t)$: conditional density of y_t at its true quantile q_t ; its inverse is called the “sparsity”
- ▶ few observations near the quantile (small f_t) mean an imprecise $\hat{\beta}$; f_t is estimated with a kernel and a bandwidth
- Time series: quantile autoregression [Koenker and Xiao \(2006\)](#) lets AR coefficients vary across quantiles; CAViaR makes the quantile itself autoregressive

CAViaR: the four specifications



Robert Engle, 2022

Photo: Jon Demske (2022); CC BY-SA

4.0; Wikimedia Commons

- **Engle and Manganelli (2004)**, written for $\text{VaR}_t = -q_t(\alpha) > 0$; y_{t-1} : yesterday's return; β_j : parameters:
 - **SAV** (symmetric absolute value): $\text{VaR}_t = \beta_1 + \beta_2 \text{VaR}_{t-1} + \beta_3 |y_{t-1}|$
 - ▶ β_2 : the persistence of the tail; β_3 : the reaction to the size of yesterday's return
 - **AS** (asymmetric slope): $\text{VaR}_t = \beta_1 + \beta_2 \text{VaR}_{t-1} + \beta_3 (y_{t-1})^+ + \beta_4 (y_{t-1})^-$
 - ▶ $(y)^+ = \max(y, 0)$, $(y)^- = -\min(y, 0)$: gains and losses enter separately; $\beta_4 > \beta_3$ is a leverage effect
- **IG** (indirect GARCH): $\text{VaR}_t = (\beta_1 + \beta_2 \text{VaR}_{t-1}^2 + \beta_3 y_{t-1}^2)^{1/2}$, a GARCH(1,1) with i.i.d. innovations
- **Adaptive**: $\text{VaR}_t = \text{VaR}_{t-1} + \beta_1 ([1 + e^{G(y_{t-1} + \text{VaR}_{t-1})}]^{-1} - \alpha)$
 - ▶ the bracket is a smooth hit indicator ($G = 10$): VaR rises after a hit and falls slowly otherwise; no distribution is assumed in any of the four

Estimation and inference (1/2): the criterion

- Regression-quantile criterion: the average pinball loss of the implied quantile path

$$\hat{\beta} = \arg \min_{\beta} \frac{1}{T} \sum_t \rho_{\alpha}(y_t + \text{VaR}_t(\beta))$$

- ▶ $\text{VaR}_t(\beta)$: the path generated by the recursion with parameters β ; the criterion is non-differentiable and non-convex in β
- EM procedure (empirical section)
 - ▶ VaR_1 : the empirical quantile of the first 300 days; 10^4 random parameter vectors, the 10 best refined by alternating simplex and quasi-Newton steps

Estimation and inference (2/2): the covariance

- Consistency and asymptotic normality, with a sandwich covariance

$$\sqrt{T}(\hat{\beta} - \beta) \rightarrow N(0, \alpha(1 - \alpha)D^{-1}AD^{-1}), \quad A = \mathbb{E}[\nabla q_t \nabla q_t'], \quad D = \mathbb{E}[f_t(q_t) \nabla q_t \nabla q_t']$$

- $\nabla q_t = \partial q_t(\beta) / \partial \beta$: gradient of the quantile path, which follows its own recursion (or numerical differentiation of the path)
 - $f_t(q_t)$: conditional density of y_t at the quantile, as in quantile regression

- Estimator of D : count the residuals within $\hat{c}$ of zero

$$\hat{D} = \frac{1}{2T\hat{c}} \sum_t 1\{|y_t - \hat{q}_t| < \hat{c}\} \nabla \hat{q}_t \nabla \hat{q}_t'$$

- $\hat{c}$: bandwidth, from the Hall–Sheather rule on the residual scale [Koenker \(2005\)](#); the dynamic model is a recursive M-estimator

The dynamic quantile (DQ) test

- The hit sequence is a martingale difference under correct specification

$$\text{Hit}_t = 1\{y_t < q_t\} - \alpha, \quad \mathbb{E}[\text{Hit}_t \mid \mathcal{F}_{t-1}] = 0$$

- ▶ regress Hit_t on $X_t = (1, \text{Hit}_{t-1}, \dots, \text{Hit}_{t-4}, q_t)$, all in $\mathcal{F}_{t-1}$ [Engle and Manganelli \(2004\)](#)

- Out-of-sample statistic: a Wald test that all regression coefficients are zero

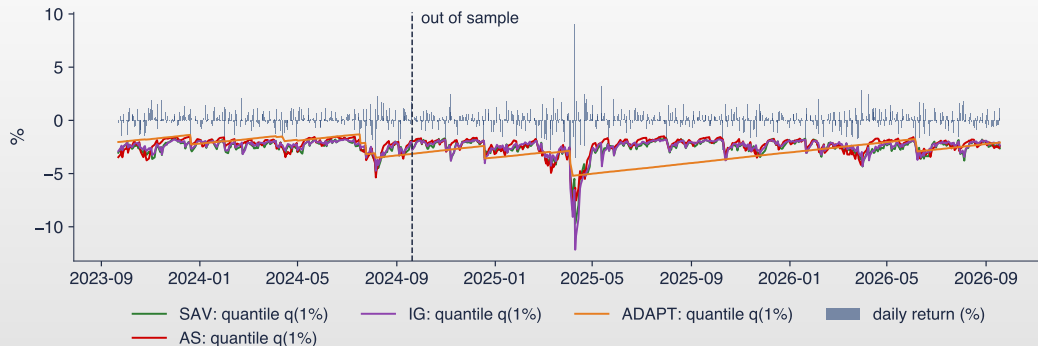
$$\text{DQ} = \frac{\text{Hit}'X(X'X)^{-1}X'\text{Hit}}{\alpha(1-\alpha)} \rightarrow \chi_6^2$$

- ▶ Hit: vector of hits; X : matrix with rows X_t ; 6 = number of regressors; a large DQ rejects calibration
- ▶ in sample the estimated β makes the hits orthogonal to the gradients: the covariance needs the correction derived by EM
- Kupiec (constant only) and Christoffersen (the first lag of the hits) are special cases; the VaR regressor adds power against level-dependent misspecification
- With $\alpha = 1\%$ and a few hundred days the χ^2 approximation is poor (Section 5): report simulated or exact p-values when possible

Case study: Engle and Manganelli (2004) on today's data

- ▣ The design of the empirical section of EM: 3,392 daily returns, the first 2,892 for estimation, the last 500 out of sample; VaR 1% and 5%
 - ▶ EM used 1986–1999 (GM, IBM, S&P 500); we use the S&P 500 from 25 March 2013 to 18 September 2026, out of sample from 20 September 2024
- ▣ Same specifications, starting values and search; the April 2025 tariff shock falls in the out-of-sample period
- ▣ Questions: which specification fits the 1% tail, are the forecasts calibrated out of sample, and is the response to bad news asymmetric?

CAViaR VaR 1% of the S&P 500



- The last year of the estimation sample and the 500 out-of-sample days; lines: the 1% return quantile $q_t = -\text{VaR}_t$ of the four specifications

Estimates, standard errors and DQ tests

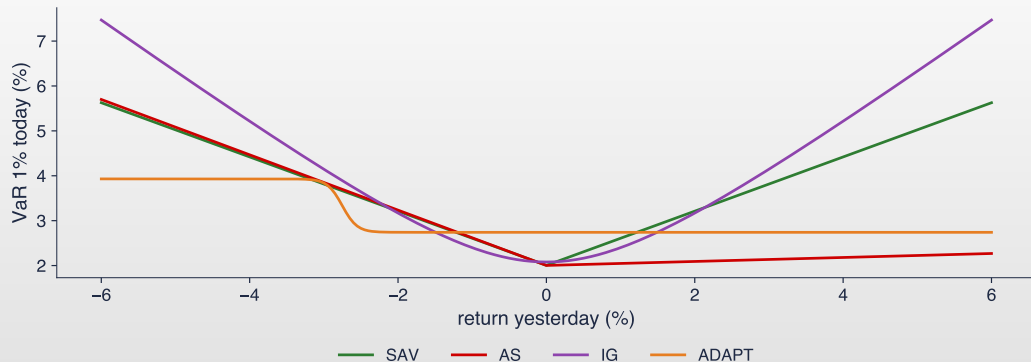
Model	β_1	β_2	β_3	β_4	$RQ \times 10^2$	hits in sample (%)	hits out of sample (%)	DQ out of sample (p)
SAV	0.359 (0.217)	0.702 (0.118)	0.604 (0.243)	—	3.305	1.0	1.4	14.4 (0.025)
AS	0.264 (0.137)	0.805 (0.080)	0.044 (0.185)	0.616 (0.149)	3.220	1.0	1.2	15.8 (0.015)
IG	0.942 (0.467)	0.674 (0.105)	1.429 (0.702)	—	3.267	1.0	1.2	2.3 (0.891)
ADAPT	1.190 (0.069)	—	—	—	4.044	0.9	0.8	27.5 (<0.001)

- Standard errors from the asymptotic covariance of EM; RQ: the minimised in-sample criterion; DQ out of sample with four lagged hits and the VaR, χ_6^2 ; 500 days give 5 expected hits
- Out-of-sample average pinball loss ($\times 10^2$): SAV 3.87, AS 3.66, IG 3.66, adaptive 4.42

Interpreting the CAViaR estimates

- All three autoregressive models fit the in-sample rate (about 1%) by construction: the quantile loss calibrates the level
- AS: a negative return moves VaR about 14 times more than a positive one of the same size; β_3 is not distinguishable from zero
- Out of sample IG passes DQ ($p = 0.891$); SAV ($p = 0.025$) and AS ($p = 0.015$) are rejected at 5%, the adaptive model clearly ($p < 0.001$)
- Five expected hits in 500 days: rejections come from clustered hits in April 2025, not from the count; such evidence is fragile
- The adaptive model reacts only to hits, like a slow thermostat: after the April 2025 cluster it stays too high for months

News impact curves



□ VaR_t as a function of y_{t-1} , with VaR_{t-1} at its in-sample median (2.16% for AS); estimated parameters of the table

Interpreting the news impact curves

- ▣ SAV and IG are symmetric by design: a 5% rally raises the 1% VaR as much as a 5% fall
- ▣ AS is almost flat for gains and steep for losses: the leverage effect estimated directly in the tail
- ▣ The adaptive curve is a step: only the sign of the hit matters, not its size
- ▣ The shape of the curve is a testable restriction: AS nests SAV ($\beta_3 = \beta_4$), a Wald test with the asymptotic covariance of EM

Recap: CAViaR

- ▣ Quantile regression is M-estimation with the pinball loss; its covariance needs the density at the quantile
- ▣ CAViaR makes the quantile autoregressive; estimation needs many starts
- ▣ DQ tests hits against information; the asymmetric model fits equity tails

M-estimation with the FZ0 loss

- A dynamic model for the pair, estimated by minimising the average FZ0 loss [Patton, Ziegel and Chen \(2019\)](#)

$$(v_t, e_t) = (v, e)(\mathcal{F}_{t-1}; \theta), \quad \hat{\theta} = \arg \min_{\theta} \frac{1}{T} \sum_t L_{\text{FZ0}}(y_t, v_t(\theta), e_t(\theta); \alpha)$$

- ▶ θ : parameters of the two recursions; consistency follows from the strict consistency of FZ0 and the identification of θ
- ▶ asymptotic normality as for CAViaR, with a sandwich covariance
- Why not maximum likelihood?
 - ▶ MLE needs the whole distribution; a wrong body or right tail distorts the left tail estimates
 - ▶ FZ estimation targets exactly the two tail functionals: less efficient if the model is right, robust if it is not (PZC, Tables 3–4)
- Related: the asymmetric Laplace quasi-likelihood [Taylor \(2019\)](#); ES regressions [Dimitriadis and Bayer \(2019\)](#); score-driven (GAS) dynamics [Creal, Koopman and Lucas \(2013\)](#)

The two-factor GAS model

- Patton, Ziegel and Chen (2019), eq. (9)–(16): the pair follows a VAR(1) driven by the scaled score of FZ0 (generalised autoregressive score, GAS)

$$\begin{pmatrix} v_{t+1} \\ e_{t+1} \end{pmatrix} = w + B \begin{pmatrix} v_t \\ e_t \end{pmatrix} + A \lambda_t$$

- w : intercept vector; B : diagonal 2×2 persistence matrix; A : 2×2 matrix of reactions to the forcing variables $\lambda_t = (\lambda_{v,t}, \lambda_{e,t})'$
- Forcing variables from the FZ0 score

$$\lambda_{v,t} = -v_t(1\{y_t \leq v_t\} - \alpha), \quad \lambda_{e,t} = \frac{1}{\alpha} 1\{y_t \leq v_t\} y_t - e_t$$

- both are identification functions: zero conditional mean under correct specification
- the Hessian is scaled with $f_t(v_t) \approx k_\alpha/v_t$, k_α a constant (exact for location–scale returns), so no density is estimated
- eight parameters; on non-hit days $\lambda_{e,t} = -e_t$ and both risk measures decay deterministically towards their means

One-factor, GARCH-FZ and Hybrid models

- **GAS-1F**, eq. (20): one latent log-scale κ_t multiplies both risk measures

$$v_t = a e^{\kappa_t}, \quad e_t = b e^{\kappa_t}, \quad \kappa_t = \beta \kappa_{t-1} + \gamma \frac{-1}{e_{t-1}} \left(\frac{1}{\alpha} \mathbf{1}_{\{y_{t-1} \leq v_{t-1}\}} y_{t-1} - e_{t-1} \right)$$

- ▶ $b < a < 0$: VaR and ES at $\kappa_t = 0$; β : persistence of the log-scale; γ : reaction to the ES forcing variable
- ▶ the intercept ω is not identified together with (a, b) and is fixed at 0

- **GARCH-FZ**, eq. (25)–(26): $\kappa_t^2 = 1 + \beta \kappa_{t-1}^2 + \gamma y_{t-1}^2$, $(v_t, e_t) = (a, b) \kappa_t$: a GARCH whose scale κ_t is fitted to the tail
- **Hybrid**, eq. (27): GAS-1F plus $\delta \ln |y_{t-1}|$ in the recursion of κ_t
 - ▶ reacts every day through δ , more strongly on hit days
- In all three the ratio $e_t/v_t = b/a$ is constant: the shape of the tail is fixed, only its scale moves

Estimating non-smooth recursive models

- ▣ The FZ0 objective is discontinuous in θ (through $1\{y_t \leq v_t(\theta)\}$) and the recursion propagates every jump
- ▣ Appendix C of the paper [Patton, Ziegel and Chen \(2019\)](#): replace the indicator by a logistic approximation in the loss and in the forcing variable

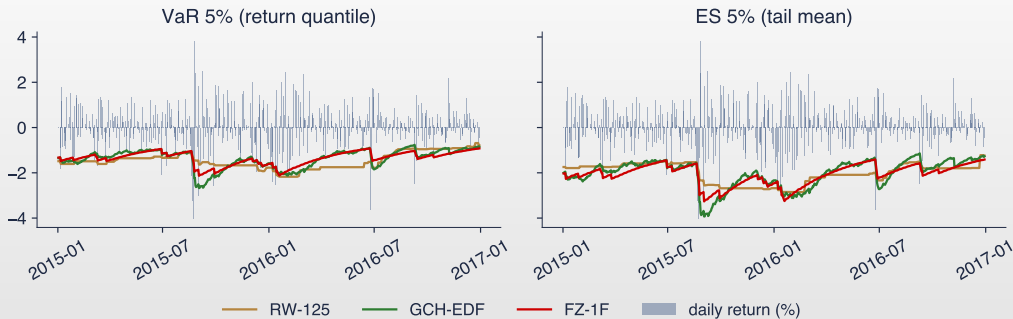
$$\Gamma(y, v; \tau) = \frac{1}{1 + e^{\tau(y-v)}}$$

- ▶ $\tau > 0$: smoothing parameter; as $\tau \rightarrow \infty$, $\Gamma \rightarrow 1\{y \leq v\}$
- ▶ quasi-Newton with $\tau = 5$, then $\tau = 20$, then the exact loss with the simplex method, each step starting from the previous one
- ▣ Our starting values: random dynamic parameters with (a, b) matched to the in-sample VaR and ES; the four best are refined
- ▣ Constraints enforced by penalty: $e_t \leq v_t$, $e_t < 0$, stationarity of the recursion

Case study: Patton, Ziegel and Chen (2019), Section 5

- ▣ Data: daily S&P 500 returns, January 1990 – December 2016; estimation on the first ten years, parameters kept fixed for 2000–2016 ($T = 4\,275$)
- ▣ Ten models, as in the paper
 - ▶ rolling windows of 125, 250 and 500 days (RW); ARMA (order by BIC) – GARCH(1,1) with Normal, skewed-t or empirical innovations
 - ▶ GAS-2F, GAS-1F, GARCH-FZ and Hybrid estimated by FZ0 minimisation
- ▣ Outputs replicated: average out-of-sample FZ0 losses (Table 8, $\alpha = 5\%$; Table S5, $\alpha = 2.5\%$), DM statistics (Table 9), goodness-of-fit tests (eq. 41)
- ▣ Our data: EODHD closes of the same index; differences of a few thousandths are expected from data revisions and optimiser paths

Three ways to forecast the tail

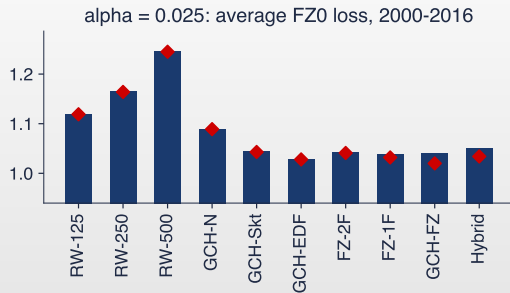
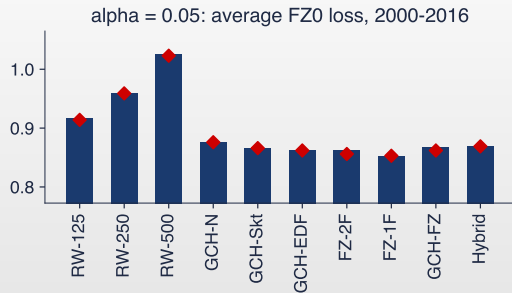


□ VaR and ES 5% of the S&P 500 in 2015–2016 (PZC, Figure 5 on our data): RW-125, GARCH-EDF and GAS-1F

Interpreting the three forecasts

- ▣ The rolling window moves in steps as extreme days enter and leave the window, months after the shock
- ▣ GARCH-EDF moves every day with squared returns; GAS-1F moves only on hit days and then decays smoothly
- ▣ August 2015: GARCH reacts most, GAS-1F jumps less but stays; in 2008 the GAS-1F ES 5% reached -9.9%
- ▣ Which reaction is right is decided by the FZ0 loss, not by the look of the path

Replication: average out-of-sample FZ0 loss



◆ Patton, Ziegel and Chen (2019) ■ our data (EODHD), same design

□ Bars: our data and code, same design; diamonds: the values printed in PZC, Table 8 ($\alpha = 5\%$) and Table S5 ($\alpha = 2.5\%$)

Replication in numbers

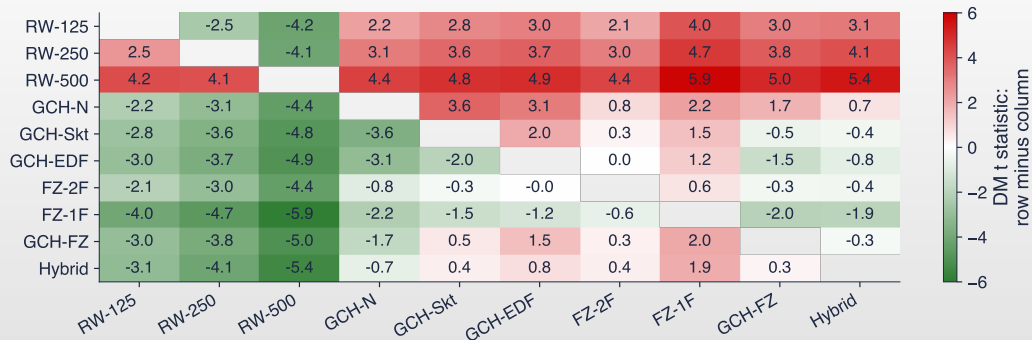
Model	$\alpha = 5\%$		$\alpha = 2.5\%$		GoF $p, \alpha = 5\%$	
	ours	PZC	ours	PZC	VaR	ES
RW-125	0.916	0.914	1.120	1.119	0.004	0.005
RW-250	0.959	0.959	1.165	1.164	<0.001	0.011
RW-500	1.025	1.023	1.247	1.245	<0.001	<0.001
GCH-N	0.876	0.876	1.089	1.089	0.018	<0.001
GCH-Skt	0.865	0.866	1.044	1.043	0.001	0.001
GCH-EDF	0.862	0.862	1.028	1.028	0.001	0.007
FZ-2F	0.862	0.856	1.044	1.041	<0.001	0.003
FZ-1F	0.853	0.853	1.038	1.032	0.001	0.037
GCH-FZ	0.867	0.862	1.040	1.020	0.005	0.016
Hybrid	0.869	0.869	1.051	1.034	<0.001	<0.001

- Largest gap: 0.006 at 5%, 0.020 at 2.5%; BIC picks ARMA(0, 0) for the mean (PZC: ARMA(1,1)); GARCH $\alpha = 0.052$, $\beta = 0.942$

Interpreting the replication

- ▣ The ranking of the paper is reproduced: GAS-1F lowest at 5% (0.853, paper 0.853), RW-500 worst, GARCH-N behind the other GARCH models
- ▣ GAS-1F estimates: $\beta = 0.995$, $\gamma = -0.0069$ (PZC, Table 7: 0.990 and -0.010); Hybrid $\delta = 0.017$ (0.018)
- ▣ Not reproduced: the goodness-of-fit tests; PZC find GAS-1F passing (p 0.242 and 0.313), on our data it is rejected (0.001; 0.037)
- ▣ Average losses are robust to small data differences; tests on a handful of tail days are not, and the covariance choice of the regression matters

Diebold–Mariano tests on FZ0 losses



- S&P 500, 2000–2016, $\alpha = 5\%$: DM statistic of row minus column, Newey–West variance; red: the row model is worse (PZC, Table 9)

Interpreting the DM matrix

- ▣ Column GAS-1F, ours against PZC: RW-125 4.02 (3.91), RW-500 5.87 (5.48), GARCH-N 2.22 (1.99), GARCH-EDF 1.19 (1.20)
- ▣ The worst models are easily separated; the best few are not (GARCH-Skt, GARCH-EDF, FZ-2F against GAS-1F below 1.96)
- ▣ One difference: GARCH-FZ against GAS-1F, 2.02 here and 1.27 in the paper; the scale normalisation of GARCH-FZ makes its estimates fragile
- ▣ Ninety pairwise tests need a multiple-testing answer: the MCS (Section 6)

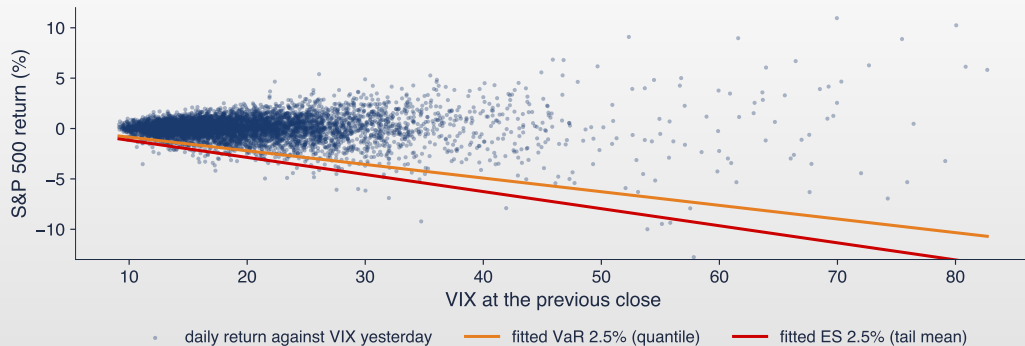
Joint (VaR, ES) regression

- Dimitriadis and Bayer (2019): both functionals are linear in the covariates

$$q_{\alpha}(Y_t | x_t) = x_t' \beta, \quad -ES_{\alpha}(Y_t | x_t) = x_t' \gamma$$

- ▶ x_t : covariates known at $t - 1$; β, γ : the effects on the quantile and on the tail mean; estimated jointly by minimising a Fissler–Ziegel loss
- ▶ M-estimator: consistent and asymptotically normal for any member of the class; the choice of (G_1, G_2) affects only efficiency
- Uses
 - ▶ which variables move the tail and by how much: the ES analogue of quantile regression
 - ▶ ES backtests: regress realised returns on the ES forecast and test intercept 0, slope 1 [Bayer and Dimitriadis \(2022\)](#) (MFM, Chapter 8)
- Application: S&P 500 returns 2000–2026 on the VIX of the previous close, $\alpha = 2.5\%$, FZ0 loss, moving-block bootstrap standard errors (blocks of 20 days)

The tail of the S&P 500 against the VIX



- Daily returns against the VIX of the previous close, $T = 6714$; lines: fitted 2.5% quantile and tail mean of the joint regression

Interpreting the joint regression

- ▣ Quantile: $0.499 - 0.135 \text{ VIX}$ (s.e. 0.150; 0.009); tail mean: $0.524 - 0.170 \text{ VIX}$ (s.e. 0.245; 0.015)
- ▣ One VIX point lowers the 2.5% quantile by 0.14 and the tail mean by 0.17 percentage points: the tail fans out 1.25 times faster in ES
- ▣ The quantile part is close to quantile regression (0.560, -0.139); in-sample hit rate 2.50%; the PZC tests do not reject ($p = 0.978$; 0.744)
- ▣ The VIX is a market forecast of volatility: the regression says that the option market prices the left tail almost linearly

Recap: Semiparametric (VaR, ES) models

- ▣ FZ0 minimisation estimates the tail pair without a distribution
- ▣ GAS forcing variables are identification functions: the model corrects itself only when the tail speaks
- ▣ The PZC ranking replicates; their goodness-of-fit results do not; joint regression links the tail to covariates

Recap: counting hits

- Hits $I_t = 1\{y_t < v_t\}$; a correct VaR makes I_t i.i.d. Bernoulli(α) (MFM, Chapter 8)
 - ▶ Kupiec (1995): unconditional coverage, likelihood-ratio statistic $LR \sim \chi_1^2$ (is the hit rate α ?)
 - ▶ Christoffersen (1998): independence of consecutive hits, and conditional coverage (both together), χ_2^2
 - ▶ Basel traffic light Basel Committee (1996): green up to 4 hits of VaR 1% in 250 days, yellow 5–9, red from 10
- What counting misses
 - ▶ hits that depend on information other than the last hit (DQ)
 - ▶ the size of tail losses (ES backtests); the spacing of hits (duration tests)
 - ▶ the effect of estimated parameters on the null distribution

Conditional calibration

- A forecast x_t of T is **conditionally calibrated** if its identification function V has conditional mean zero [Nolde and Ziegel \(2017\)](#)

$$\mathbb{E}[V(x_t, Y_t) \mid \mathcal{F}_{t-1}] = 0 \Rightarrow \mathbb{E}[h_{t-1} V(x_t, Y_t)] = 0$$

- ▶ h_{t-1} : any instrument known at $t - 1$ (constant, lagged hits, the forecast itself); a moment test (Wald, χ^2)
- PZC, eq. (40)–(41): standardised generalised residuals

$$\lambda_{v,t}^s = 1\{y_t \leq v_t\} - \alpha, \quad \lambda_{e,t}^s = \frac{1\{y_t \leq v_t\} y_t}{\alpha e_t} - 1$$

- ▶ regress each on (1, its own first lag, v_t or e_t) and test that all three coefficients are zero; we use a White (HC0) covariance
- ▶ DQ is the special case for VaR; the ES regression tests ES only jointly with VaR, as elicibility predicts
- Power comes from the instruments: lagged hits detect clustering, the forecast level detects scale errors

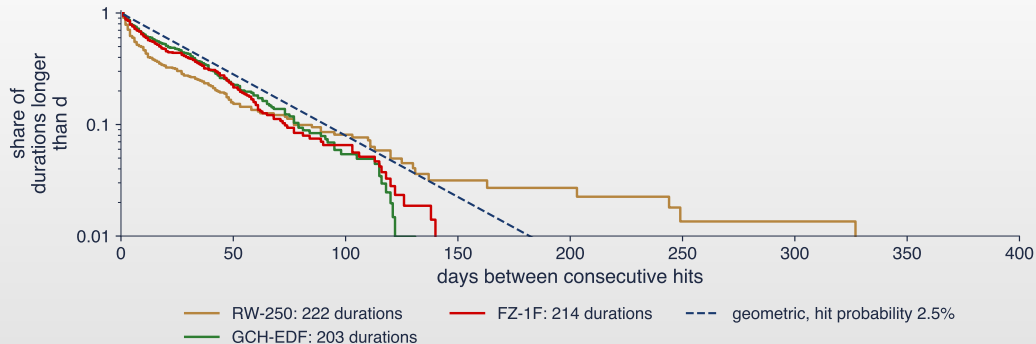
Duration-based backtests

- ▣ Christoffersen and Pelletier (2004): under a correct VaR the durations d_i (days between consecutive hits) are geometric, memoryless, with mean $1/\alpha$
 - ▶ alternative: Weibull hazard $\lambda(d) = a^b b d^{b-1}$, the probability of a hit after d quiet days; $a > 0$: scale, $b > 0$: shape
 - ▶ $b = 1$: constant hazard (no memory, as under H_0); $b < 1$: decreasing hazard, i.e. hits cluster
- ▣ Likelihood with censoring: the first and last spells are incomplete and enter through the survival function $S(d) = e^{-(ad)^b}$

$$\text{LR} = 2[\ln L(\hat{a}, \hat{b}) - \ln L(\tilde{a}, 1)] \rightarrow \chi_1^2$$

- ▶ $(\hat{a}, \hat{b})$: unrestricted estimates; $\tilde{a}$: estimate under $b = 1$; small-sample p-values by simulation are advisable
- ▣ Durations see clustering at any distance, not only on consecutive days as Christoffersen's test does

How far apart are the hits?



- S&P 500, VaR 2.5%, out of sample 2000–2026; empirical survival of the durations between hits (log scale) against the geometric law of a correct model

Interpreting the durations

- ▣ RW-250: 40% of the durations are at most 5 days (geometric: 12%); Weibull $b = 0.67$, $p < 0.001$: strong clustering
- ▣ GARCH-EDF: $b = 0.92$, $p = 0.107$; GAS-1F: $b = 0.88$, $p = 0.019$: dynamic models remove most of the clustering
- ▣ The long tail of RW-250 (calm years without hits) is the other face of clustering: the window remembers old crises
- ▣ All models have too many hits overall (Section 5 table): parameters fixed in 1999 are too optimistic after 2000

Backtesting ES: what each test needs

Test	Statistic	Inputs
McNeil and Frey (2000)	mean of $(y_t - e_t)/\sigma_t$ on hit days, bootstrap	v_t, e_t, σ_t
Acerbi and Székely (2014) Z_2	$1 - \frac{1}{T\alpha} \sum_t \frac{y_t l_t}{e_t}$	v_t, e_t and the forecast distribution (simulated p -value)
Du and Escanciano (2017)	cumulative violations $H_t = \frac{1}{\alpha}(\alpha - u_t)\mathbf{1}\{u_t \leq \alpha\}$	the PIT $u_t = F_{t t-1}(y_t)$
Kratz, Lok and McNeil (2018) Patton, Ziegel and Chen (2019), Bayer and Dimitriadis (2022)	multinomial test of several VaR levels regression on (v_t, e_t)	the PIT, or VaR at several levels v_t, e_t

- σ_t : forecast volatility; l_t : hit indicator; $u_t = F_{t|t-1}(y_t)$: probability integral transform (PIT), uniform on $[0, 1]$ under a correct forecast distribution
- Only the regression tests use nothing beyond the pair (VaR, ES): the others are tests of a distribution, which ES alone does not supply

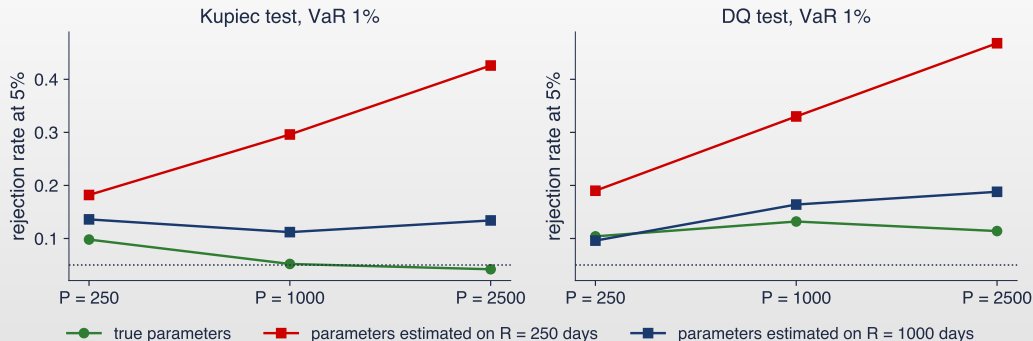
Estimation risk in backtests

- Backtests assume the VaR is known; in practice $\hat{v}_t = v_t(\hat{\theta}_R)$, estimated on R days and tested on the next P days [Escanciano and Olmo \(2010\)](#)

$$\frac{1}{\sqrt{P}} \sum_t (\hat{l}_t - \alpha) = \frac{1}{\sqrt{P}} \sum_t (l_t - \alpha) + \underbrace{\sqrt{P/R} \cdot \mathbb{E}[f_t(v_t) \nabla v_t'] \sqrt{R}(\hat{\theta}_R - \theta)}_{\text{estimation term}}$$

- ▶ $\hat{l}_t, l_t$: hits of the estimated and of the true VaR; $f_t(v_t)$: conditional density at the VaR; ∇v_t : gradient of the VaR in θ
- The estimation term vanishes only if $P/R \rightarrow 0$; with a fixed window it grows with the test sample
 - ▶ the Kupiec variance $\alpha(1 - \alpha)$ is then too small and the test over-rejects a correct model
 - ▶ fixes: the corrected variance of Escanciano and Olmo, a subsampling or bootstrap of the whole estimate-and-test procedure
- A Monte Carlo shows the size of the problem for the windows banks actually use

Size of backtests with estimated parameters



- GARCH(1,1) data ($\omega = 0.02$, $\alpha = 0.08$, $\beta = 0.90$, Normal), the correct model estimated by QML on R days, VaR 1% on the next P days; 500 replications; nominal size 5%

Interpreting the Monte Carlo

- ▣ Known parameters: Kupiec rejects 10%, 5% and 4% for $P = 250, 1000, 2500$: close to 5% except with few hits
- ▣ Estimated on $R = 250$: 18%, 30%, 43%: a correct model is rejected more often the longer we test
- ▣ With $R = 1000$ the distortion shrinks (13% at $P = 2500$); the hit rate across replications has s.d. 0.99 pp against 0.29 pp
- ▣ DQ is oversized even with known parameters (10%): rare hits make the χ^2_6 approximation poor

Calibration tests across assets



- PZC goodness-of-fit regressions for VaR and ES 2.5%, ten models, parameters estimated on the first ten years (five for Bitcoin) and kept fixed to 18 September 2026

Interpreting the calibration tests

- Only 4 of 50 asset–model pairs pass both regressions at 10%: over 10–26 years with fixed parameters almost every model is miscalibrated somewhere
- S&P 500 hit rates at 2.5%: GAS-1F 3.2%, GARCH-EDF 3.0%, RW-250 3.3%: the level drifts after the estimation decade
- EUR/RON: estimated on 2005–2015 (crisis and float), the GARCH models over-predict risk afterwards (GARCH-EDF 0.6% hits): miscalibration in the safe direction is rejected too
- Bitcoin: short history and noisy tail, low power; the simple models pass (RW-500 2.5%)
- A backtest answers “is it calibrated?”; whether to prefer one failing model to another is a comparison question

Recap: Backtesting

- Backtests are moment tests of the identification function with chosen instruments
- Durations catch clustering at any distance; ES tests need more than ES, except the regression tests
- Estimated parameters inflate the size; long, fixed-parameter evaluations reject almost everything

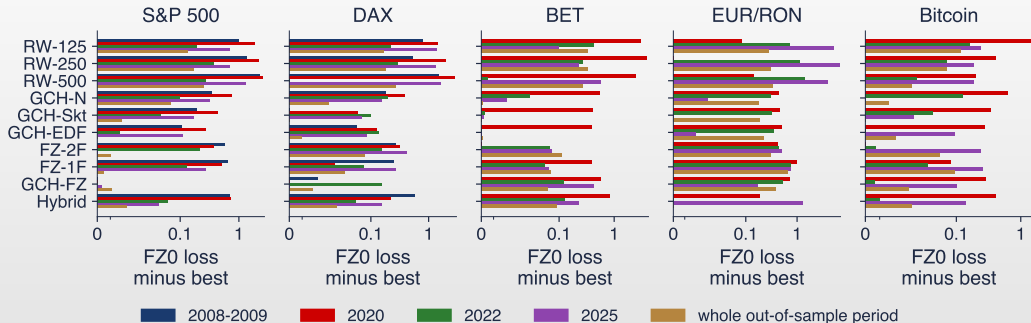
From backtests to comparative backtests

- [Nolde and Ziegel \(2017\)](#): a traditional backtest tests H_0 : “the bank’s model is calibrated”; a comparative backtest tests the bank’s model against a standard one with a consistent score
 - ▶ H_0^- : internal at least as good as standard; H_0^+ : at most as good; green if H_0^+ is rejected, red if H_0^- is rejected, yellow otherwise
- Information sets matter [Holzmann and Eulert \(2014\)](#): a calibrated forecast based on more information has a lower expected consistent score
 - ▶ so a ranking by FZ0 rewards both calibration and information, which a backtest cannot do
- Tools from [Chapter 1](#): DM with HAC variance [Diebold and Mariano \(1995\)](#), Giacomini–White for estimated forecasting methods [Giacomini and White \(2006\)](#), the MCS [Hansen, Lunde and Nason \(2011\)](#)

Design of the comparison

- ▣ Ten PZC models, $\alpha = 2.5\%$, FZ0 losses out of sample: S&P 500 and DAX from 2000, BET from 2010, EUR/RON from July 2015, Bitcoin from 2020
- ▣ 90% MCS with the $T_{\max}$ statistic, moving-block bootstrap (blocks of 10 days over the whole period, 5 days in a stress period), 1000 replications
- ▣ Stress periods fixed before looking at the losses
 - ▶ 2008: September 2008 – March 2009; 2020: 19 February – 30 June 2020; 2022: the calendar year; 2025: March – June 2025
 - ▶ S&P 500 in 2020: 93 days, of which 9 GAS-1F hits; in 2025: 83 days, 5 hits
- ▣ A stress window holds a few tail days: expect wide MCS sets and unstable winners

Losses in stress periods



- Average FZ0 loss of each model minus that of the best model in the same period (symmetric log scale); bars missing when the period is not out of sample

Interpreting the stress-period losses

- Winners over the whole period: S&P 500 GCH-EDF, DAX GCH-Skt, BET GCH-Skt, EUR/RON Hybrid, Bitcoin GCH-Skt
- In 2020 the winner changes: S&P 500 GCH-FZ, BET FZ-2F, Bitcoin FZ-2F; rolling windows lose most in every crash
- Crashes separate models by an order of magnitude more than calm years: an average over the whole period is dominated by a few weeks
- No model wins every crisis; the semiparametric models are competitive but not dominant out of sample after 2016

Model confidence sets



□ MCS p -values (FZ0 loss, $\alpha = 2.5\%$); green: in the 90% MCS

Interpreting the MCS

- ▣ Whole period, MCS sizes: S&P 500 6, DAX 6, BET 3, EUR/RON 8, Bitcoin 10 of 10
- ▣ Rolling windows are eliminated on the long equity samples; in 2020 and 2022 almost everything survives (2020: 10 on the S&P 500; 2022: 10)
- ▣ EUR/RON in 2020 is the exception: only 4 models survive, the rolling windows and the Hybrid
- ▣ A large MCS is a result, not a failure: the crisis windows are too short to separate tail forecasts

Recap: Comparison

- ▣ Comparative backtests put the burden of proof on the model, and reward information
- ▣ Rankings change in crises; whole-period averages hide this
- ▣ Report the MCS, not the winner

Why the square-root-of-time rule fails (1/2)

- The rule scales the one-day VaR to h days

$$\text{VaR}^{(h)} = \sqrt{h} \text{VaR}^{(1)}$$

- ▶ $\text{VaR}^{(h)}$: VaR of the h -day return $\sum_{k=1}^h y_{t+k}$; exact for i.i.d. returns from the Normal distribution with zero mean
- ▶ for i.i.d. stable laws with index $\gamma \in (0, 2]$ the factor is $h^{1/\gamma}$ ($\gamma = 2$: the Normal case)
- Fat tails that are not stable: the h -day sum is closer to Normal than the daily return
 - ▶ so $\sqrt{h}$ overstates the quantile at long horizons [Danielsson and Zigrand \(2006\)](#)
- Leverage (GJR) makes multi-day losses more skewed than daily ones: the left tail of the sum is thicker than the scaled daily tail

Why the square-root-of-time rule fails (2/2)

- Volatility clustering: for GARCH(1,1) the variance of the h -day return depends on today's volatility

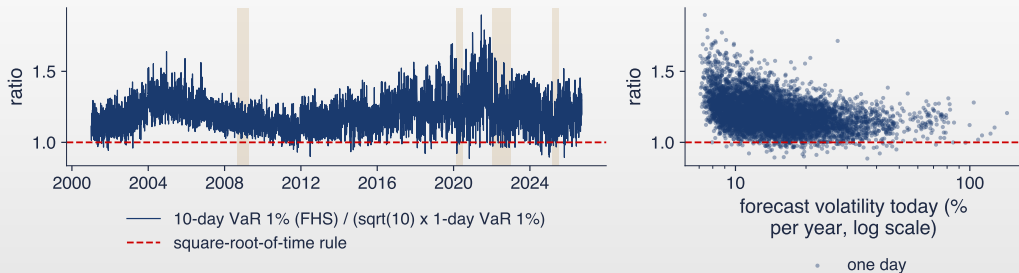
$$\text{Var}_t\left(\sum_{k=1}^h y_{t+k}\right) = \sum_{k=0}^{h-1} [\bar{\sigma}^2 + (\alpha + \beta)^k (\sigma_{t+1}^2 - \bar{\sigma}^2)]$$

- ▶ $\bar{\sigma}^2 = \omega / (1 - \alpha - \beta)$: unconditional variance; σ_{t+1}^2 : tomorrow's forecast variance; $\alpha + \beta$: persistence
- ▶ in calm times ($\sigma_{t+1}^2 < \bar{\sigma}^2$) $\sqrt{h}$ understates risk, in crises it overstates it: mean reversion of volatility
- Remedies: simulate the h -day distribution (FHS), or forecast the h -day quantile directly (quantile regression on h -day returns)

Multi-day forecasts and their backtests

- ▣ FHS: GJR-GARCH(1,1) by QML Bollerslev (1986), Glosten, Jagannathan and Runkle (1993)
 - ▶ rolling 2000-day window, re-estimated every 250 days
 - ▶ 2000 paths of 10 days with bootstrapped standardised residuals
- ▣ Backtesting 10-day VaR
 - ▶ daily 10-day forecasts overlap: hits are MA(9) even under H_0 ; counting them as independent inflates the size (Chapter 0, overlapping observations)
 - ▶ options: non-overlapping windows (every 10th day, few hits), or HAC variance on overlapping hits
- ▣ Basel uses 10-day horizons and liquidity-adjusted horizons for ES (MAR33): the horizon problem is regulatory, not academic

10-day VaR 1% against the square-root-of-time rule



- S&P 500, 2001–2026: ratio of the 10-day VaR 1% by FHS to $\sqrt{10}$ times the 1-day VaR 1% of the same model; right: the ratio against today's forecast volatility

Interpreting the horizon ratio

- ▣ Median ratio 1.19, range 0.89 to 1.90: $\sqrt{10}$ scaling usually understates the 10-day VaR 1% of the S&P 500
- ▣ Low-volatility days: median 1.26; high-volatility days: 1.14: mean reversion pushes the ratio towards 1 in crises
- ▣ Non-overlapping backtest (646 windows, 6.5 expected hits): FHS 6, $\sqrt{10}$ rule 7; Kupiec p 0.85 and 0.83
- ▣ So few hits cannot separate the two: horizon errors of this size are invisible to a non-overlapping backtest

Recap: The horizon

- The square-root rule needs i.i.d. Normal returns; clustering, leverage and fat tails break it in opposite directions
- Simulate the horizon or model it directly; backtest with non-overlapping windows or HAC variances
- Multi-day tests have very little power

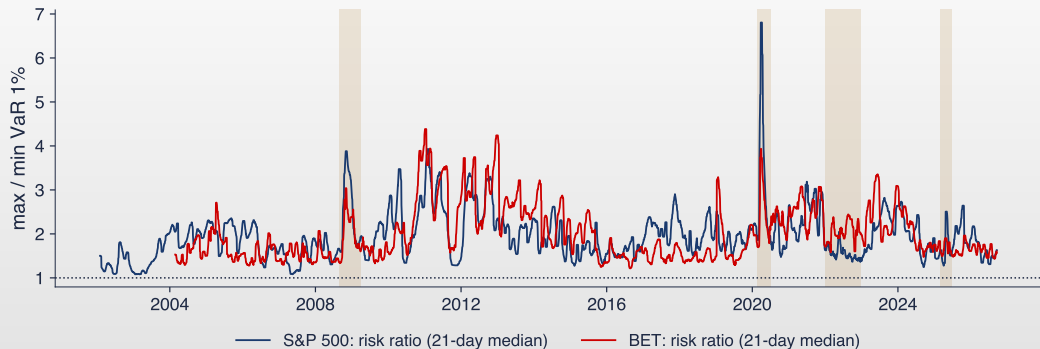
Model risk of risk models

- Danielsson et al. (2016): the **risk ratio** across standard models measures the disagreement a regulator should expect

$$RR_t = \frac{\max_m VaR_{m,t}}{\min_m VaR_{m,t}} \geq 1$$

- ▶ $VaR_{m,t}$: the forecast of model m for day t ; $RR_t = 1$: all models agree; 2: the most prudent model asks for twice the capital of the least prudent
- ▶ they find the ratio largest exactly in crises, when risk numbers matter most
- Boucher et al. (2014): **risk models-at-risk**: quantify the model risk of a VaR forecast and adjust the forecast for estimation and specification errors
- Kellner and Röscher (2016): ES carries more model risk than VaR at comparable levels: it extrapolates further into the tail
- Our six models, rolling 1000-day windows: historical simulation, Normal with window volatility, EWMA ($\lambda = 0.94$), GARCH-N, GARCH-t, filtered HS

How much do standard models disagree?



- Risk ratio of six VaR 1% forecasts, 21-day rolling median, S&P 500 and BET, 2002–2026 (BET from 2004); shaded: stress periods

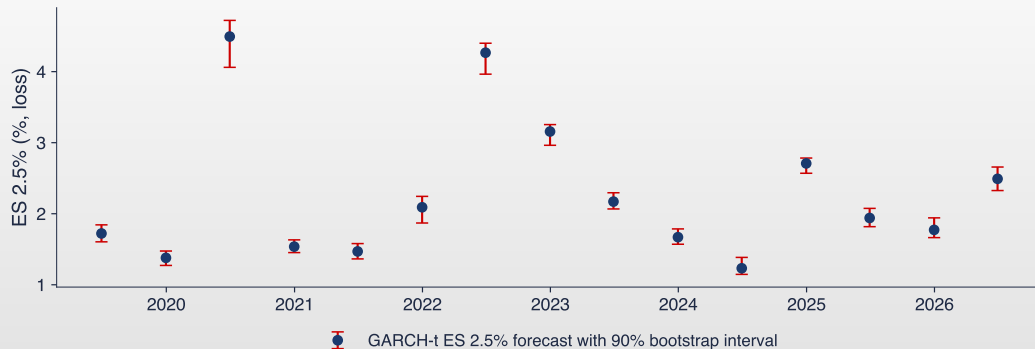
Interpreting the risk ratio

- Median risk ratio 1.9 (S&P 500) and 1.8 (BET); 90th percentile 2.7 and 3.0; maximum 10.5 on 17 March 2020
- Spring 2020: median 2.9 (S&P 500), 2.5 (BET): the slow historical simulation and the fast EWMA diverge most after a jump
- HS is the most conservative model on 73% of S&P 500 days, EWMA the least on 53%
- Hit rates for VaR 1% (S&P 500): HS 1.4%, GARCH-N 2.2%, GARCH-t 1.4%, FHS 1.3%: the disagreement is not noise, the Normal models are wrong

Estimation risk in tail forecasts

- Gao and Song (2008): for GARCH VaR and ES, $\sqrt{T}(\widehat{ES}_t - ES_t)$ is asymptotically normal; the variance has a volatility part and a tail-shape part
- Bootstrap Christoffersen and Gonçalves (2005): re-estimate the model on simulated paths and recompute the forecast **conditional on the observed history**
 - ▶ algorithm: fit GARCH-t on 2000 days; simulate 150 paths of 2000 days from the fit; re-estimate on each; filter the observed returns with each $\hat{\theta}^*$; ES^* for tomorrow
 - ▶ 90% interval: the 5% and 95% quantiles of ES^*
- The interval measures parameter uncertainty only: it is conditional on the model being right, so it is a lower bound on total uncertainty

An ES forecast with its estimation error



- S&P 500, GARCH-t ES 2.5% for the next day at half-year ends 2019–2026, with 90% parametric bootstrap intervals (150 re-estimations each)

Interpreting the estimation error

- ▣ Interval width relative to the forecast: median 13%, from 8% to 19%
- ▣ June 2020: ES 4.49% with interval [4.06; 4.72]: two thousand days of data still leave a band of several tenths of a percent
- ▣ Estimation risk is small next to model risk (risk ratios near 2): the choice of model matters more than its precision
- ▣ Capital rules that multiply a point forecast ignore both: reporting the interval is the honest minimum

Recap: Model and estimation risk

- Standard models disagree by a factor of about two, more in crises
- Bootstrap intervals for ES are narrow next to that disagreement
- Report both: the forecast, its interval and the spread across models

Extremes of dependent series

- For a stationary series, dependence changes the law of the maximum only through one number, the **extremal index** θ [Embrechts, Klüppelberg and Mikosch \(1997\)](#)

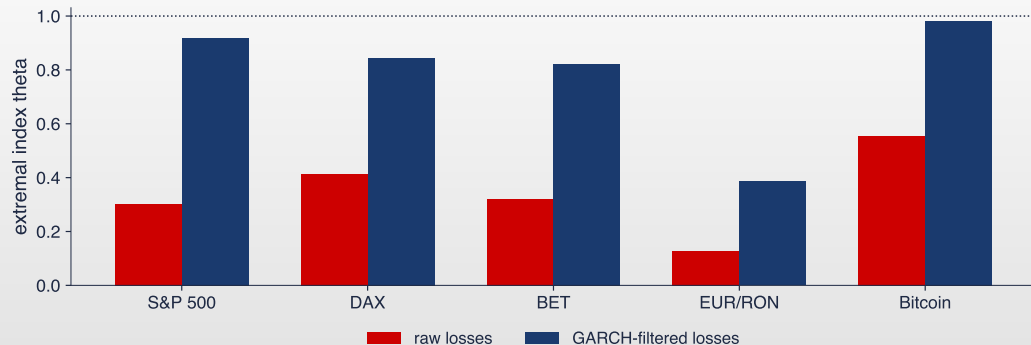
$$\Pr \left(\max_{t \leq n} X_t \leq u_n \right) \approx F(u_n)^{n\theta}, \quad \theta \in (0, 1]$$

- ▶ X_t : the daily loss; F : its marginal distribution; u_n : a high threshold that grows with the sample size n
- ▶ $1/\theta$ = mean cluster size of exceedances; $\theta = 1$: no clustering of extremes
- Intervals estimator [Ferro and Segers \(2003\)](#), from the gaps T_i between the N exceedances (when $\max T_i > 2$)

$$\hat{\theta} = \min \left(1, \frac{2(\sum_i (T_i - 1))^2}{(N - 1) \sum_i (T_i - 1)(T_i - 2)} \right)$$

- [McNeil and Frey \(2000\)](#): fit GARCH, apply EVT to the standardised residuals; it works if filtering removes the clustering (θ near 1 after filtering)

Extremal index before and after filtering



- Exceedances of daily losses over their 95% quantile, whole samples; raw losses and GARCH-standardised losses (in-sample ARMA-GARCH parameters)

Interpreting the extremal index

- ▣ Raw losses: $\hat{\theta} = 0.30$ (S&P 500), 0.32 (BET), 0.13 (EUR/RON): extremes come in clusters of two to eight days
- ▣ After filtering: 0.92, 0.85, 0.98 for the S&P 500, DAX and Bitcoin: GARCH removes most of the clustering, as McNeil and Frey assume
- ▣ EUR/RON stays clustered (0.39) and BET partly (0.82): a managed exchange rate and a thin market have dependence that GARCH does not capture
- ▣ With fixed in-sample parameters the filter is imperfect after the estimation period: part of the remaining clustering is drift

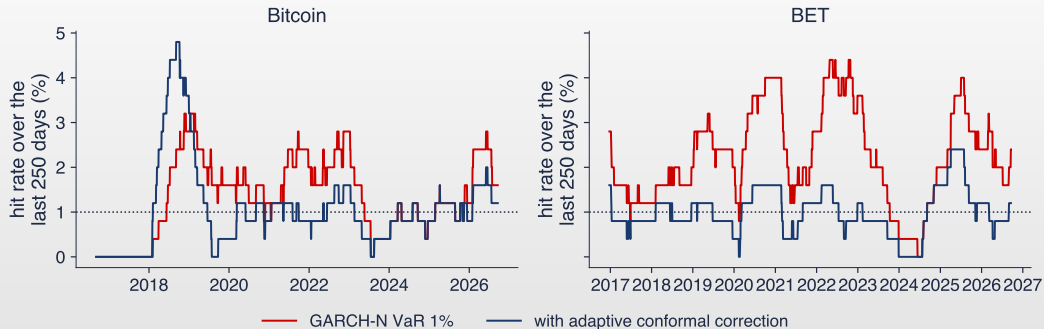
Conformal calibration of VaR

- Split conformal quantiles guarantee coverage under exchangeability ([Chapter 13](#)); returns are not exchangeable, so the guarantee fails
- **Adaptive conformal inference** (ACI) [Gibbs and Candès \(2021\)](#): forecast at a working level α_t and correct it after each day

$$\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t), \quad \text{err}_t = 1\{y_t < q_t(\alpha_t)\}$$

- ▶ $q_t(\alpha_t)$: the model's quantile at level α_t ; $\gamma > 0$: step size; after a hit α_{t+1} falls (VaR rises), after a quiet day it rises slightly
- ▶ deterministic guarantee: $\left| T^{-1} \sum_t \text{err}_t - \alpha \right| \leq \frac{\max(\alpha_1, 1 - \alpha_1) + \gamma}{\gamma T}$ for **any** sequence of returns (α_1 : the starting level)
- ▶ long-run frequency only: no conditional calibration, no statement about ES
- Further reading: conformal recalibration of extreme tail quantiles under dependence [Pele et al. \(Conformal_Oracle\)](#)

Adaptive conformal correction of VaR 1%



- GARCH-N VaR 1% (rolling 1000-day window, re-estimated every 250 days) from 2016, and its ACI correction with $\gamma = 0.005$; rolling 250-day hit rates

Interpreting the conformal correction

- BET: GARCH-N hits 2.36% of 2 675 days (Kupiec $p < 0.001$); with ACI 1.08% ($p = 0.666$), conditional coverage $p = 0.662$
- Bitcoin: 1.28% to 1.02%; the rolling hit rate is not smoothed: its maximum rises from 3.2% to 4.8% after over-correction
- ACI fixes the frequency, not the dynamics: the adapted level can go below zero (VaR then infinite) after a run of quiet days
- A useful wrapper for a regulator's count; a calibrated model is still needed for DQ and for ES

Recap: Extremes and conformal calibration

- ▣ Extremes cluster; filtering removes most clustering for liquid markets, not for managed rates
- ▣ ACI guarantees the long-run hit frequency for any data, nothing more
- ▣ Both are corrections of a model, not replacements for one

An open question

- Do semiparametric (VaR, ES) models beat location–scale models out of sample once estimation is repeated, and is the advantage concentrated in crises?
 - ▶ formal: H_0 : equal expected FZ0 loss of GAS-1F and GARCH-EDF with rolling re-estimation, on pre-registered assets, periods and levels
 - ▶ falsified by a significant GW statistic in the pre-registered crisis windows, robust across $\alpha = 1\%$, 2.5% , 5%
- Why it matters: Basel requires ES models; the PZC evidence comes from fixed parameters and four indices
 - ▶ literature to start from: Patton, Ziegel and Chen (2019), Taylor (2019), Nolde and Ziegel (2017), Dimitriadis and Bayer (2019)

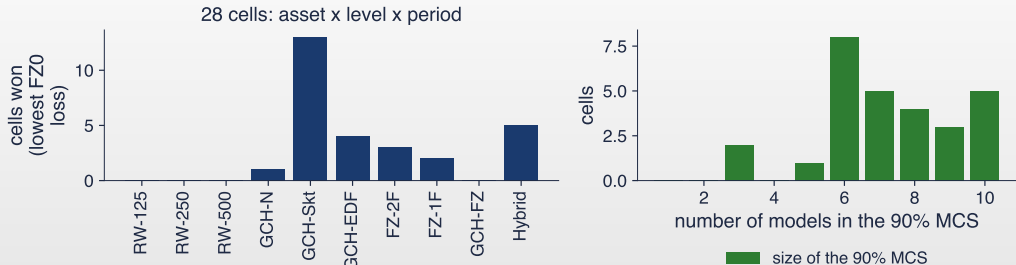
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
 - ▶ **literature:** List peer-reviewed papers since 2017 that compare dynamic ES models out of sample with FZ losses; give DOIs. Then check every DOI on Crossref
 - ▶ **hypothesis:** Under which data-generating processes should a one-factor GAS model beat GARCH with empirical innovations for ES 2.5%?
 - ▶ **code and replication:** ask for the GAS-1F recursion, then reproduce a published number first (PZC, Table 8: 0.853)
 - ▶ **critique:** Act as a hostile referee: list the ways an FZ0 ranking could be an artefact of the sample, the level or the starting values.
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

What the human checks

- ▣ Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- ▣ The loss is consistent for the target: no “average ES error”, no MSE of VaR forecasts
- ▣ The convention is right: VaR 1% is minus the 1% quantile of returns, never “VaR 99%”; signs of v and e are consistent
- ▣ Forecasts use only information available at $t - 1$; the estimation window and re-estimation scheme are stated
- ▣ Rankings are reported with DM or MCS uncertainty and checked on a Murphy diagram

Mini-case: is there a best ES model?



- 28 cells: five assets $\times$ two levels (2.5%, 5%) $\times$ up to three periods (whole, before 2020, from 2020); left: the model with the lowest FZ0 loss; right: size of the 90% MCS
- 6 different winners; GARCH-Skt wins 13 cells, GAS-1F 2; MCS sizes from 3 to 10, median 7; GAS-1F is in the MCS in 22 cells: an AI summary that names “the best ES model” is wrong

Project idea

- ▣ **ES models for Central and Eastern European markets under rolling re-estimation:** replicate first, then extend
 - ▶ replicate: PZC Table 8 and Table S5 for the S&P 500 (GAS-1F 0.853 and 1.032) and the EM design with VaR 1%
 - ▶ extend: BET, WIG20, BUX, PX and EUR/RON; rolling re-estimation; joint (VaR, ES) regressions with VIX and the BNR rate; ACI as a wrapper
 - ▶ pre-register: samples, levels, models, windows, the stress periods and the MCS design
- ▣ Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- ▣ A risk forecast is a point forecast of a functional; judge it with a loss consistent for that functional
- ▣ VaR is elicitable, ES only with VaR; FZ0 is the scale-free choice for time series
- ▣ CAViaR and GAS-FZ models estimate the tail directly; their published rankings replicate, their published tests do not
- ▣ Backtests are moment tests with estimation risk; comparisons need DM, MCS and Murphy diagrams
- ▣ Horizon, model and estimation risk are larger than most reported confidence suggests

Self-assessment

Questions

- Why does a non-convex level set rule out a strictly consistent loss for ES?
- Which choice of (G_1, G_2) gives FZ0, and what does zero homogeneity buy?
- Why is the DQ test with a VaR regressor more powerful than Christoffersen's test?
- Why does a fixed estimation window inflate the size of the Kupiec test?
- When does $\sqrt{10}$ scaling overstate the 10-day VaR?

Next:

- Long memory and rough volatility
- ARFIMA, fractional cointegration, rough volatility

Appendix: consistency of GPL quantile scores

- The expected GPL score of a report x under $Y \sim F$

$$\mathbb{E}_F S(x, Y) = \int_{-\infty}^x (1 - \alpha)(G(x) - G(y)) dF(y) + \int_x^{\infty} \alpha(G(y) - G(x)) dF(y)$$

- Differentiate (Leibniz; the boundary terms vanish)

$$\frac{d}{dx} \mathbb{E}_F S(x, Y) = G'(x) [(1 - \alpha)F(x) - \alpha(1 - F(x))] = G'(x)(F(x) - \alpha)$$

- $G' \geq 0$: the derivative is ≤ 0 for $x < q_\alpha$ and ≥ 0 for $x > q_\alpha$, so q_α minimises; strictly if G is strictly increasing and F has a unique α -quantile
- The same computation with $G(x) = x$ is the first-order condition of quantile regression

Appendix: FZ0, consistency and homogeneity

- Fix e ; in v , FZ0 is $-\frac{1}{\alpha e} [1\{y \leq v\}(v - y) - \alpha v]$ plus terms free of v : since $-1/(\alpha e) > 0$, a positive multiple of the pinball loss, minimised at q_α
- At $v = q_\alpha$: $\mathbb{E} L = \frac{c}{e} + \ln(-e) - 1$ with $c = v - \frac{1}{\alpha} \mathbb{E}[1\{Y \leq v\}(v - Y)] = \mathbb{E}[Y | Y \leq q_\alpha] = -\text{ES}_\alpha$; the derivative $-c/e^2 + 1/e$ vanishes at $e = c$
- Homogeneity: replacing (y, v, e) by (sy, sv, se) , $s > 0$, changes FZ0 only by $\ln s$, the same for every forecast: loss differences are scale-free
- Joint minimisation over (v, e) with $e \leq v < 0$ therefore returns $(q_\alpha, -\text{ES}_\alpha)$

References (1/4)

- ▣ Acerbi, C., & Székely, B. (2014). Backtesting expected shortfall. *Risk*, December 2014 (MSCI working paper). [msci.com](https://www.msci.com)
- ▣ Acerbi, C., & Tasche, D. (2002). On the coherence of expected shortfall. *Journal of Banking & Finance*, 26(7), 1487–1503. doi:10.1016/S0378-4266(02)00283-2
- ▣ Artzner, P., Delbaen, F., Eber, J.-M., & Heath, D. (1999). Coherent measures of risk. *Mathematical Finance*, 9(3), 203–228. doi:10.1111/1467-9965.00068
- ▣ Basel Committee on Banking Supervision (2019). *MAR33: Internal models approach, capital requirements calculation*. Basel Framework, Bank for International Settlements. [bis.org](https://www.bis.org)
- ▣ Basel Committee on Banking Supervision (1996). *Supervisory framework for the use of “backtesting in conjunction with the internal models approach to market risk capital requirements*. Bank for International Settlements. [bis.org](https://www.bis.org)
- ▣ Bayer, S., & Dimitriadis, T. (2022). Regression-based expected shortfall backtesting. *Journal of Financial Econometrics*, 20(3), 437–471. doi:10.1093/jjfinec/nbaa013
- ▣ Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327. doi:10.1016/0304-4076(86)90063-1
- ▣ Boucher, C. M., Daniélsson, J., Kouontchou, P. S., & Maillet, B. B. (2014). Risk models-at-risk. *Journal of Banking & Finance*, 44, 72–92. doi:10.1016/j.jbankfin.2014.03.019
- ▣ Christoffersen, P. F. (1998). Evaluating interval forecasts. *International Economic Review*, 39(4), 841–862. doi:10.2307/2527341
- ▣ Christoffersen, P., & Gonçalves, S. (2005). Estimation risk in financial risk management. *The Journal of Risk*, 7(3), 1–28. doi:10.21314/JOR.2005.112
- ▣ Christoffersen, P., & Pelletier, D. (2004). Backtesting value-at-risk: A duration-based approach. *Journal of Financial Econometrics*, 2(1), 84–108. doi:10.1093/jjfinec/nbh004
- ▣ Creal, D., Koopman, S. J., & Lucas, A. (2013). Generalized autoregressive score models with applications. *Journal of Applied Econometrics*, 28(5), 777–795. doi:10.1002/jae.1279
- ▣ Daniélsson, J., & Zigrand, J.-P. (2006). On time-scaling of risk and the square-root-of-time rule. *Journal of Banking & Finance*, 30(10), 2701–2713. doi:10.1016/j.jbankfin.2005.10.002

References (2/4)

- Danielsson, J., James, K. R., Valenzuela, M., & Zer, I. (2016). Model risk of risk models. *Journal of Financial Stability*, 23, 79–91. doi:10.1016/j.jfs.2016.02.002
- Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263. doi:10.1080/07350015.1995.10524599
- Dimitriadis, T., & Bayer, S. (2019). A joint quantile and expected shortfall regression framework. *Electronic Journal of Statistics*, 13(1), 1823–1871. doi:10.1214/19-EJS1560
- Du, Z., & Escanciano, J. C. (2017). Backtesting expected shortfall: Accounting for tail risk. *Management Science*, 63(4), 940–958. doi:10.1287/mnsc.2015.2342
- Ehm, W., Gneiting, T., Jordan, A., & Krüger, F. (2016). Of quantiles and expectiles: Consistent scoring functions, Choquet representations and forecast rankings. *Journal of the Royal Statistical Society: Series B*, 78(3), 505–562. doi:10.1111/rssb.12154
- Embrechts, P., Klüppelberg, C., & Mikosch, T. (1997). *Modelling Extremal Events for Insurance and Finance*. Springer. doi:10.1007/978-3-642-33483-2
- Engle, R. F., & Manganelli, S. (2004). CAViaR: Conditional autoregressive value at risk by regression quantiles. *Journal of Business & Economic Statistics*, 22(4), 367–381. doi:10.1198/073500104000000370
- Escanciano, J. C., & Olmo, J. (2010). Backtesting parametric value-at-risk with estimation risk. *Journal of Business & Economic Statistics*, 28(1), 36–51. doi:10.1198/jbes.2009.07063
- Ferro, C. A. T., & Segers, J. (2003). Inference for clusters of extreme values. *Journal of the Royal Statistical Society: Series B*, 65(2), 545–556. doi:10.1111/1467-9868.00401
- Fissler, T., Ziegel, J. F., & Gneiting, T. (2016). Expected shortfall is jointly elicitable with value at risk: Implications for backtesting. *Risk*, January 2016, 58–61; arXiv:1507.00244. [arxiv.org](https://arxiv.org/abs/1507.00244)
- Fissler, T., & Ziegel, J. F. (2016). Higher order elicibility and Osband's principle. *The Annals of Statistics*, 44(4), 1680–1707. doi:10.1214/16-AOS1439
- Gaglianone, W. P., Lima, L. R., Linton, O., & Smith, D. R. (2011). Evaluating value-at-risk models via quantile regression. *Journal of Business & Economic Statistics*, 29(1), 150–160. doi:10.1198/jbes.2010.07318
- Gao, F., & Song, F. (2008). Estimation risk in GARCH VaR and ES estimates. *Econometric Theory*, 24(5), 1404–1424. doi:10.1017/S0266466608080559

References (3/4)

- Giacomini, R., & White, H. (2006). Tests of conditional predictive ability. *Econometrica*, 74(6), 1545–1578. doi:10.1111/j.1468-0262.2006.00718.x
- Gibbs, I., & Candès, E. (2021). Adaptive conformal inference under distribution shift. *Advances in Neural Information Processing Systems*, 34; arXiv:2106.00170. [arxiv.org](https://arxiv.org/abs/2106.00170)
- Glosten, L. R., Jagannathan, R., & Runkle, D. E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *The Journal of Finance*, 48(5), 1779–1801. doi:10.1111/j.1540-6261.1993.tb05128.x
- Gneiting, T. (2011). Making and evaluating point forecasts. *Journal of the American Statistical Association*, 106(494), 746–762. doi:10.1198/jasa.2011.r10138
- Gneiting, T., & Raftery, A. E. (2007). Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477), 359–378. doi:10.1198/016214506000001437
- Hansen, B. E. (1994). Autoregressive conditional density estimation. *International Economic Review*, 35(3), 705–730. doi:10.2307/2527081
- Hansen, P. R., Lunde, A., & Nason, J. M. (2011). The model confidence set. *Econometrica*, 79(2), 453–497. doi:10.3982/ECTA5771
- Holzmann, H., & Eulert, M. (2014). The role of the information set for forecasting – with applications to risk management. *The Annals of Applied Statistics*, 8(1), 595–621. doi:10.1214/13-AOAS709
- Kellner, R., & Rösch, D. (2016). Quantifying market risk with value-at-risk or expected shortfall? Consequences for capital requirements and model risk. *Journal of Economic Dynamics and Control*, 68, 45–63. doi:10.1016/j.jedc.2016.05.002
- Koenker, R., & Bassett, G. (1978). Regression quantiles. *Econometrica*, 46(1), 33–50. doi:10.2307/1913643
- Koenker, R. (2005). *Quantile Regression*. Cambridge University Press. doi:10.1017/CBO9780511754098
- Koenker, R., & Xiao, Z. (2006). Quantile autoregression. *Journal of the American Statistical Association*, 101(475), 980–990. doi:10.1198/016214506000000672
- Kratz, M., Lok, Y. H., & McNeil, A. J. (2018). Multinomial VaR backtests: A simple implicit approach to backtesting expected shortfall. *Journal of Banking & Finance*, 88, 393–407. doi:10.1016/j.jbankfin.2018.01.002

References (4/4)

- ▣ Kupiec, P. H. (1995). Techniques for verifying the accuracy of risk measurement models. *The Journal of Derivatives*, 3(2), 73–84. doi:10.3905/jod.1995.407942
- ▣ McNeil, A. J., & Frey, R. (2000). Estimation of tail-related risk measures for heteroscedastic financial time series: An extreme value approach. *Journal of Empirical Finance*, 7(3–4), 271–300. doi:10.1016/S0927-5398(00)00012-8
- ▣ Nolde, N., & Ziegel, J. F. (2017). Elicitability and backtesting: Perspectives for banking regulation. *The Annals of Applied Statistics*, 11(4), 1833–1874. doi:10.1214/17-AOAS1041
- ▣ Patton, A. J., Ziegel, J. F., & Chen, R. (2019). Dynamic semiparametric models for expected shortfall (and value-at-risk). *Journal of Econometrics*, 211(2), 388–413. doi:10.1016/j.jeconom.2018.10.008
- ▣ Pele, D. T., Bolovăneanu, V., Ginavar, A., Lessmann, S., & Härdle, W. K. Conformal recalibration of extreme tail quantiles under temporal dependence. Working paper and replication package (further reading). github.com
- ▣ Petropoulos, F., Apiletti, D., Assimakopoulos, V., Babai, M. Z., et al. (2022). Forecasting: Theory and practice. *International Journal of Forecasting*, 38(3), 705–871. doi:10.1016/j.ijforecast.2021.11.001
- ▣ Taylor, J. W. (2008). Estimating value at risk and expected shortfall using expectiles. *Journal of Financial Econometrics*, 6(2), 231–252. doi:10.1093/jjfinec/nbn001
- ▣ Taylor, J. W. (2019). Forecasting value at risk and expected shortfall using a semiparametric approach based on the asymmetric Laplace distribution. *Journal of Business & Economic Statistics*, 37(1), 121–133. doi:10.1080/07350015.2017.1281815
- ▣ Weber, S. (2006). Distribution-invariant risk measures, information, and dynamic consistency. *Mathematical Finance*, 16(2), 419–441. doi:10.1111/j.1467-9965.2006.00277.x
- ▣ Ziegel, J. F. (2016). Coherence and elicibility. *Mathematical Finance*, 26(4), 901–918. doi:10.1111/mafi.12080